*Earth and Planetary Science Letters*, 72 (1985) 125–140
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands



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# Strain removal applied to paleomagnetic directions in an orogenic belt: the Permian red slates of the Alpes Maritimes, France

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Received May 24, 1984
Revised version accepted October 2, 1984

Detailed investigations of relationships between strain and directions of characteristic remanent magnetization (ChRM) have been conducted on the Permian red series of the Alpes Maritimes, France. The analysis of the results obtained by structural and paleomagnetic studies on 14 sites leads to the following observations: in most sites, the mean direction of the strain shortening axis is close to the expected Permian paleomagnetic direction. In this case we observe a large scatter of the ChRM directions within each site, which is interpreted as an increase of an initial dispersion due to strain. Moreover this dispersion more or less increases with strain intensity. In other cases, where the angle between the shortening axis and the paleomagnetic direction is high, the whole population is deflected towards the cleavage ($XY$) plane. These observations lead us to assume a material line-like behaviour of the ChRM vectors and consequently an attempt has been made to recalculate these directions by strain removal. These calculations give positive results at two levels: (1) a statistically significant improvement in the clustering of the within-site distributions, and (2) after tilt correction, using unstrained bedding planes, a significant improvement in the clustering of the between-site distribution. The formation mean direction, after strain removal, is in good agreement with known Permian results for stable Europe.

## 1. Introduction

Since Runcorn [1] first demonstrated the opening of the Atlantic, using paleomagnetic data from North America and Europe, paleomagnetic methods have been widely used, mainly on a large scale, to contribute to the understanding of (1) the relative motions of the continents, and (2) apparent polar wander. More recently, attention moved towards the margins of the large plates, in order to understand the interactions between them. An increasing number of paleomagnetic studies has been done within the orogenic belts themselves, providing new constraints on tectonic evolution of mountain belts [2]. However, some special problems of deformation, metamorphism or overprinting arise in such areas, and great care is required in the interpretation of the data [3]. Up to now, the widely used "structural corrections" are in fact the familiar tilt corrections introduced by Graham [4] who compared the dispersion of data before and after rotation of the bedding plane about its strike line. More complex developments of this method have been applied for plunging fold axes [5,6] or fault rotations and have been reviewed by MacDonald [7], but all of them involve rigid rotations of blocks in which it is assumed that there is no plastic strain. However, strain has been considered as responsible for anomalous results, which show negative fold tests even though the magnetization is probably pretectonic [8] or as an alternative explanation to syntectonic remagnetization for scattered data [9]. For these reasons, penetratively deformed rocks are generally regarded as unsuitable for paleomagnetic work and very few quantitative studies have been carried out on this problem.

Nevertheless some laboratory experiments have

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been done on various synthetic samples, to study the magnetization behaviour during progressive strain. Kodama and Cox [10] did not show any coherent bias of the remanence with respect to the shortening axis; these results are thought to be due to the nature of the artificial sediment itself (kaolinite matrix) which showed a discontinuous deformation behaviour at the sample scale. On the other hand, Ozima [11] observed that "compressional plastic deformation of a Cu-Co alloy (...) resulted in a systematic rotation of remanent magnetization (IRM at H = 0.1 T) away from the axis of compression with increasing degree of the deformation". Morash and Bonhommet [12], working on hematite in a homogeneous plasticine matrix, obtained systematic rotation of the resultant IRM, the angle of rotation being close to that predicted for a macroscopic material line. Similarly, Blow and Hamilton [13] observed a significant shallowing of detrital remanent magnetization, acquired by tank redepositions of silty clay grade sediments during compactional loading. This can be attributed to vertical shortening. In this context, it would seem sensible to study the remanence behaviour in natural deformed series. The results presented here provide a tentative answer to the first question that can be asked: is there any coherent effect of plastic deformation on pretectonic remanent magnetization; in other words, does the pretectonic remanence behave like a macroscopic material line (see Appendix 1) during deformation?

To investigate this particular problem, the studied series must (a) have a pretectonic magnetization that can be recovered by magnetic analysis; (b) have good natural strain gauges in sufficient quantity.

Permian red shales and slates covering the Argentera Massif (Alpes Maritimes, France) have been chosen because they satisfy the above requirements (Fig. 1). A thick Permian sequence of sediments forms the autochthonous cover of the Argentera crystalline massif. The upper part (1 km thick) consists of red sandstones (Barrot Formation). A slight unconformity below the Werfenian quartzites points to a slight pre-Triassic deformation [14] but the main phase is of Alpine age [15]. Open to isoclinal folds show axial planar slaty cleavage. These red slates contain many ellipsoidal reduction spots and these good quality strain markers have been widely used in previous tectonic studies. Graham [16] made the first quantitative strain study in this area, revealing that strain increases from south to north. Interrelationships of deformation, metamorphism and textures were studied by Siddans [17] who concluded that illite recrystallisation during low-grade metamorphism was the predominant mechanism of cleavage development. Relationships of magnetic fabric and strain were investigated by Henry [18] and Kligfield et al. [19,20]. Paleomagnetic analysis has been done by Van Den Ende [8] within the undeformed part of the Dôme de Barrot. He determined a mean paleomagnetic direction for these series ($D = 205.5°$, $I = -13.5°$, $\alpha_{95} = 2.6°$) of Permian age and thus, the magnetization is actually pretectonic (prealpine), at least in the undeformed part of the formation.

## 2. Sampling and field measurements

Two sets of sites were sampled in the Barrot Formation (Fig. 1). The first set (group I) of nine sites (A to J) is distributed in the Dôme de Barrot [2] and along the Tinée [6] and Roya [1] valleys. Here the strain intensity increases from south to north. The other set (group II) of six sites (P1 to P6) is distributed throughout the two limbs and hinge of a 50-m-wide anticline (Fig. 2), in the Vionène valley, the most deformed part of this area (asterisk in Fig. 1c). In this fold, the finite strain intensity is roughly constant, but there are variations in the geometrical relationships of the bedding plane and principal directions of strain.

The scale of the sites was selected to ensure a good homogeneity of strain and bedding dip in each site, and was generally about 2 m × 2 m. 145 cores were drilled and oriented in-situ, and cut in the laboratory to yield 250 specimens. Field structural measurements were made at each site as follows: the axial ratios of the elliptical reduction spots were measured on at least two sets of perpendicular natural planes, as close as possible to the principal strain planes. Measurements of cleavage plane and lineations (bedding-cleavage inter-

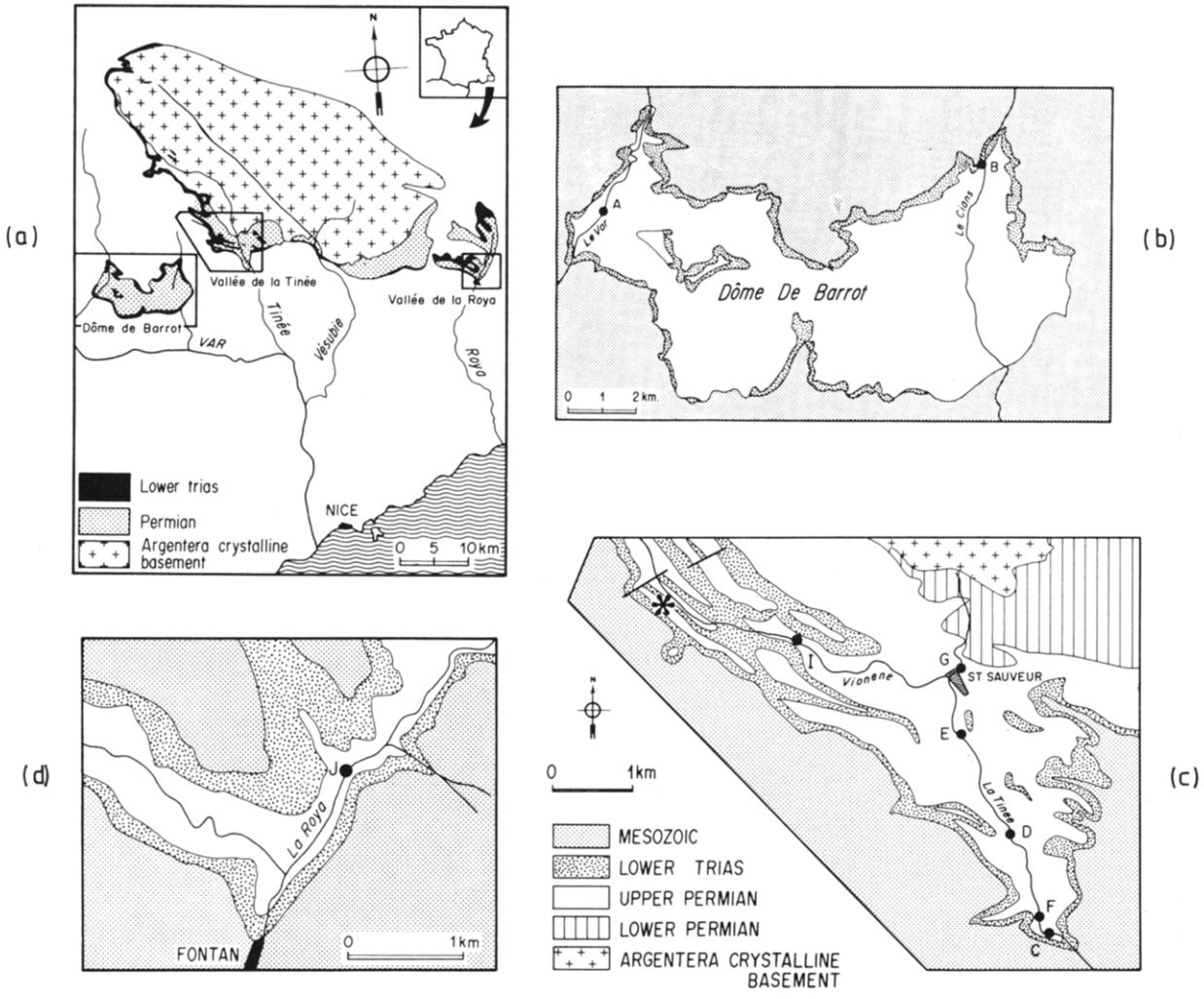



Fig. 1. (a) Location map of the main outcrops of Permian red shales and slates in the Alpes Maritimes. Sampling sites in (b) the Dôme de Barrot, (c) The Tinée and Vionène valleys, (d) the Roya valley. Asterisk in (c): group II sites.

section, stretching, etc.) give the orientation of the principal strain axes. In the most deformed sites, some difficulties arise with the determination of a bedding plane; as a matter of fact, the development of the slaty cleavage tends to erase the bedding plane and this surface may not be detectable in some times. Fortunately, the bedding trace is always marked by some green "reduction bands" on the other available planes (cleavage, joint planes, etc.). Plots of these traces on a stereogram give a mean bedding plane at every site studied.

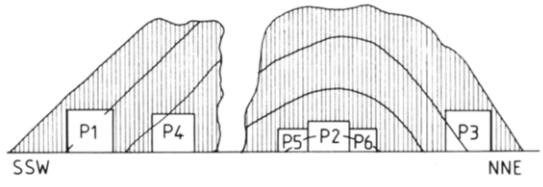



Fig. 2. Schematic cross-section through the fold in the Vionène valley showing the structural emplacements of the sites P1 to P6 (group II).

## 3. Strain analysis—results

Results of structural and strain measurements are summarized in Table 1. They allow us to determine a mean bedding plane and a mean strain tensor at each site as follows. The measurements of the axial ratios of the elliptical reduction spots are plotted on either a $R_f/\phi$ diagram [21,22] or a simple long-axis/short-axis diagram, depending on the angular fluctuations that may exist between the long axis and a reference line; this yield a mean strain ratio for each studied plane. Then, if we suppose no volume change ($\lambda_1 \cdot \lambda_2 \cdot \lambda_3 = 1$) we can calculate the elongation $\lambda_1 \geqslant \lambda_2 \geqslant \lambda_3$ along each axis. On the other hand, considering the cleavage plane as the $XY$ finite strain plane, its pole is the orientation of minimum elongation axis $Z$ and the stretching lineation is the maximum elongation axis $X$; the intermediate axis $Y$ is orthogonal to the others to form a right-handed cartesian frame.

The results obtained here (Table 1) are consistent with previous ones obtained in this area [15,16,19,20]. In the undeformed southern part of the Dôme de Barrot (site A), ellipsoids are slightly flattened in the bedding ($\lambda_1 = \lambda_2 > \lambda_3$). Towards the north, strain intensity increases (Fig. 3). In the northern part of the Dôme de Barrot (site B), although cleavage is not present, ellipsoids are prolate ($\lambda_1 > \lambda_2 = \lambda_3$) with long axes trending about N110°; in the southern Tinée valley (site C) cleavage development, superposed upon a compactional fabric, produces a pencil structure and ellipsoids are prolate, with long axes parallel to bedding/cleavage intersection; moving towards the north, ellipsoids are flattened in the slaty cleavage, first triaxial ($\lambda_1 > \lambda_2 > \lambda_3$) with $X$ parallel to the bedding/cleavage intersection (site F), then oblate ($\lambda_1 = \lambda_2 > \lambda_3$; sites D and E). Finally the ellipsoids return to a triaxial shape, with $X$ parallel to the down-dip stretching lineation within the cleavage, in the most deformed sites (G, I). The Roya valley site(J) shows an intermediate stage of this sequence, equivalent to site F. The six sites of the fold show the most deformed state, similar to those of sites G and I.

From the strike and dip of axial planar slaty cleavage we can see that the $Z$-axis is generally subhorizontal with a general trend SSW-NNE, which is more or less parallel to the Permian paleomagnetic direction of the Dôme de Barrot.

TABLE 1

Mean orientations and strain

| Sites | Bedding | | Cleavage | | $X$-axis | | Principal strains | | |
|---|---|---|---|---|---|---|---|---|---|
| | $S$ | $D$ | $S$ | $D$ | $D$ | $I$ | $\lambda_1$ | $\lambda_2$ | $\lambda_3$ |
| A | 181 | 10 | – | – | – | – | 1.193 | 1.193 | 0.702 |
| B | 242 | 5 | – | – | 305 | 5 | 1.368 | 0.855 | 0.855 |
| C | 72 | 25 | 90 | 90 | 90 | 8 | 1.478 | 0.869 | 0.778 |
| D | 105 | 30 | 285 | 64 | – | – | 1.339 | 1.339 | 0.558 |
| E | 46 | 45 | 264 | 61 | – | – | 1.546 | 1.546 | 0.418 |
| F | 96 | 37 | 102 | 90 | 102 | 4 | 1.915 | 1.063 | 0.491 |
| G | 340 | 26 | 295 | 78 | 324 | 66 | 2.629 | 1.143 | 0.346 |
| I | 317 | 26 | 295 | 67 | 353 | 13 | 2.098 | 1.499 | 0.318 |
| J | 73 | 46 | 276 | 44 | 89 | 11 | 1.643 | 1.369 | 0.444 |
| P1 | 131 | 39 | 291 | 87 | 89 | 82 | 2.072 | 1.480 | 0.326 |
| P2 | 149 | 15 | 300 | 82 | 72 | 79 | 1.789 | 1.361 | 0.410 |
| P3 | 317 | 72 | 131 | 80 | 272 | 74 | 2.002 | 1.310 | 0.381 |
| P4 | 139 | 39 | 299 | 87 | 91 | 84 | 1.894 | 1.388 | 0.380 |
| P5 | 116 | 20 | 300 | 82 | 20 | 82 | 1.789 | 1.361 | 0.410 |
| P6 | 304 | 23 | 300 | 82 | 17 | 82 | 1.789 | 1.361 | 0.410 |

$S$, $D$: strike, dip of bedding and cleavage planes, $D$, $I$: declination, inclination of $X$-axis.

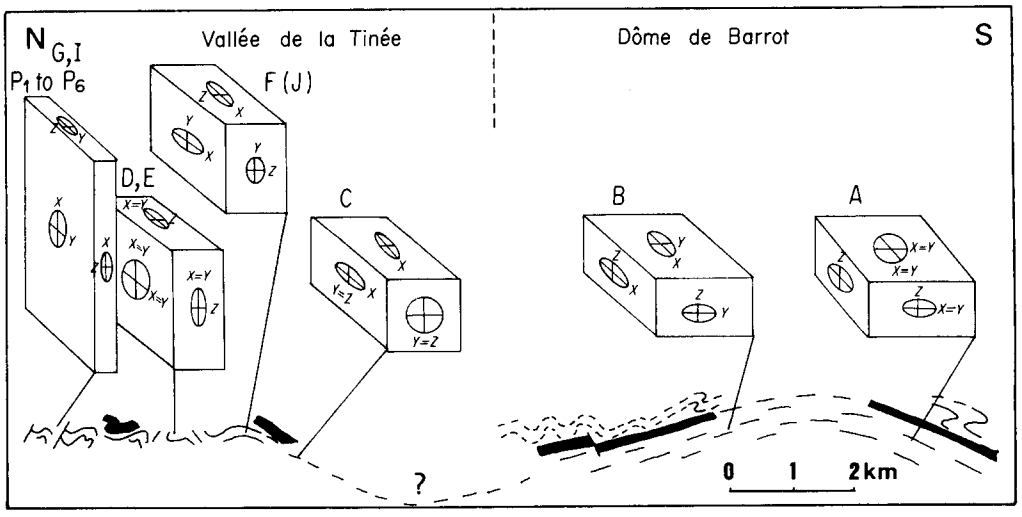



Fig. 3. Schematic cross-section through the Dôme de Barrot and the Tinée outcrops, with block-diagrams illustrating the increasing strain stages (after Graham [16]).

## 4. Paleomagnetic analysis

### *4.1. Laboratory techniques and measurements*

All the magnetic measurements have been done with a computer-assisted Schonstedt DSM-1 spinner magnetometer of sensitivity $\sim 10^{-4}$ A $m^{-1}$. Progressive isothermal remanent magnetization (IRM) acquisition experiments have been performed with a Brucker BE-10 electromagnet (maximum field 1.25 T). Thermal demagnetization procedures have been carried out with a Schonstedt TSD-1 non-magnetic furnace in a mu-metal shield with a very low residual field of about 10 nT. For chemical demagnetization, samples were prepared by drilling a 1-cm diameter hole in order to increase the leaching surface [23,24]. On the other hand, uniform leaching of the sample was helped by the presence of cleavage planes that provided good diffusivity. The samples were then immersed in HCl in individual receptacles and left in a field-free space during increasing time steps. This was performed at room temperature and required a total time ranging from 600 to 1150 hours of leaching.

### *4.2. NRM*

All 250 specimens were submitted to a short-period viscosity test [25]. 83% have a short-period viscosity coefficient lower than the 20% limit we have chosen as the maximum allowed value for our paleomagnetic investigations.

In-situ NRM directions are somewhat scattered but do not appear to show any significant present-day field overprinting (Fig. 4).

Tilt correction does not induce any improvement in these distributions either in the gradient population (group I) or in the fold (group II). So we have here an apparent paradox: on the one hand NRM directions are closed to the expected Permian direction (SW declination and shallow) and therefore should be considered as pretectonic; but on the other hand, tilt correction does not improve these distributions, which is normally characteristic of syn- to post-tectonic magnetizations. One possible reason for this is strain.

### *4.3. Thermal demagnetization*

Thermal demagnetizations were carried out on 76 specimens and have shown single (Fig. 5a) or double (Fig. 5b) component NRM behaviour. In

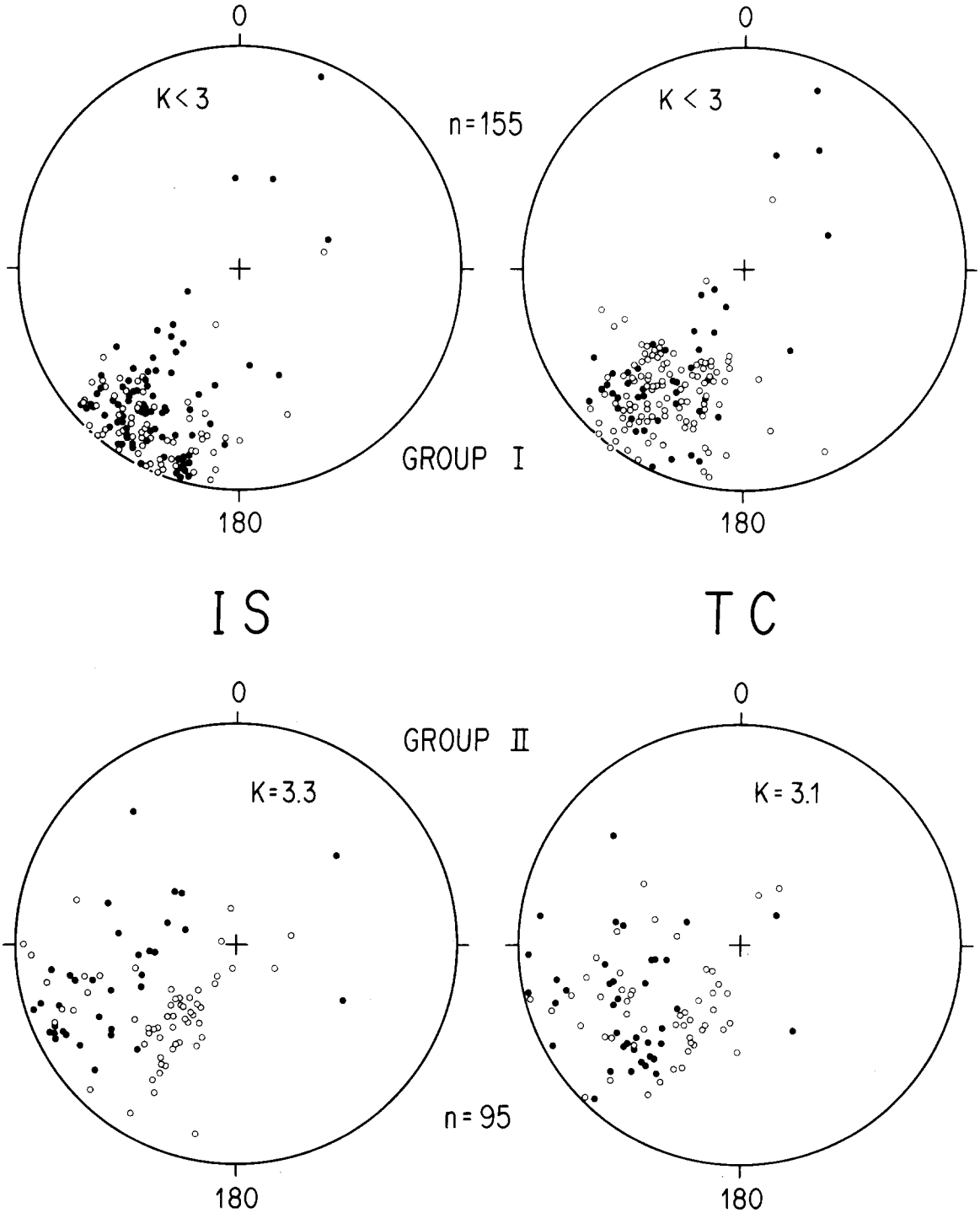



Fig. 4. NRM directions of the specimens of group I (top) and group II (bottom). IS: in-situ (left); TC: tilt-corrected (right). Open symbols: negative inclinations. Closed symbols: positive inclinations. Stereographic projections.

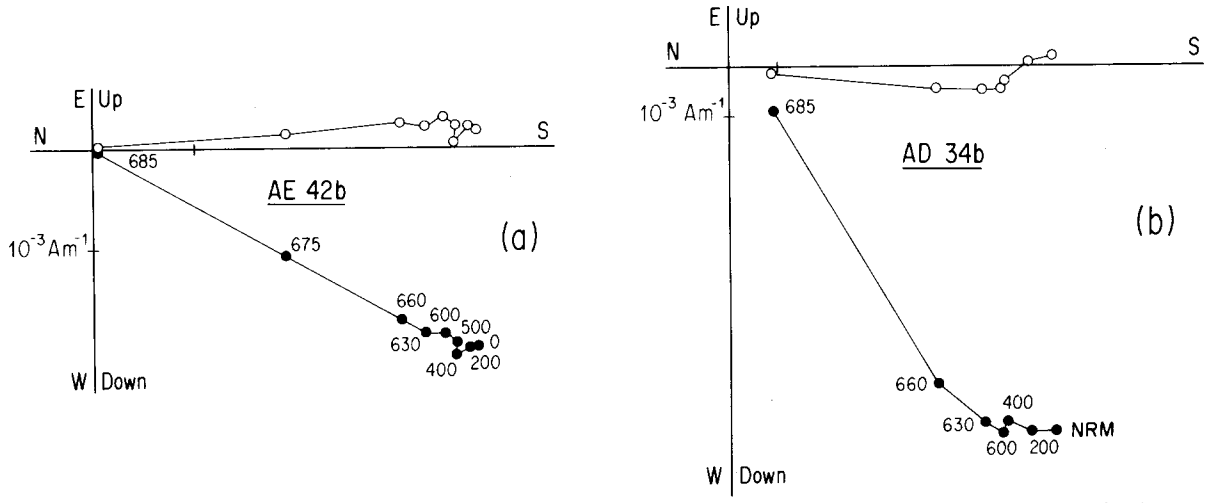



Fig. 5. Thermal demagnetization curves in Zijderveld [40] vector projections. Open symbols: projection of the vector end-points onto the vertical plane. Closed symbols: projection onto the horizontal plane. Temperature in °C is indicated.

the latter case, it was often impossible to separate the two components by this technique because of a nearly complete overlapping of the unblocking temperature spectra. This kind of data was treated by the great circle method [26]. Another problem especially present in the most deformed sites, was the appearance of strong viscous magnetization components in the higher-temperature steps of the procedure, that conceal the characteristic magnetization. Some IRM acquisitions performed on unheated and heated specimens [27] show the presence of a low-coercivity magnetic carrier (magnetic or maghemite) after annealing and an absence of this phase in the fresh specimens. Thus we appear to have magnetite production during the heating process as described by Dunlop [28]. When these mineralogical modifications occurred, we rejected the results from further consideration.

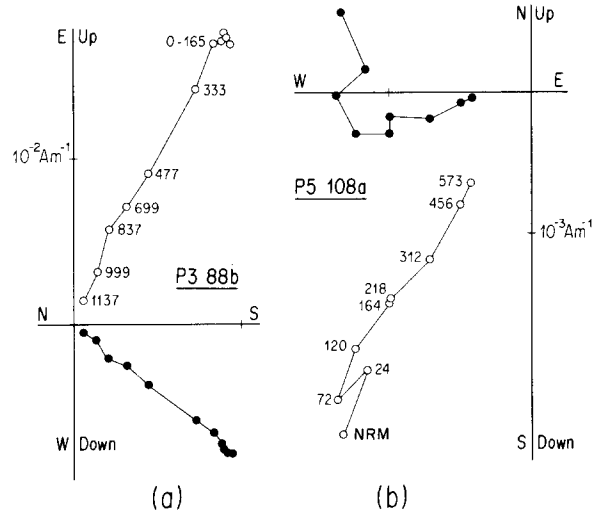



Fig. 6. Chemical demagnetization curves. Cumulative time of leaching is indicated in hours. Same conventions as in Fig. 5.

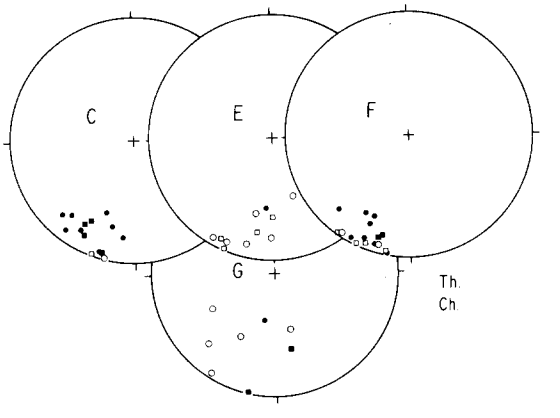



Fig. 7. In-situ ChRM directions obtained by thermal (circles) and chemical (squares) demagnetization in four sites of group I. Same conventions as in Fig. 4.

### 4.4. Chemical demagnetization

In order to isolate a characteristic component and to avoid mineralogical changes during heating, chemical demagnetization by hydrochloric acid was performed on 82 specimens. In this case, too, we observe either single (Fig. 6a) or double (Fig. 6b) component behaviour, but here the two components appear to be clearly separated during leaching. IRM acquisitions done on pairs of leached and non leached specimens [27] show that more than 90% of the magnetization is carried by a very high-coercivity phase, fine-grained hematite, so the NRM of these rocks is probably of chemical origin [29–31].

### 4.5. Summary of data obtained by demagnetization

Three kinds of magnetic behaviour have been recovered:

(1) Thermal demagnetization led to determination of remagnetization planes for each treated specimen (sites A and I). In this case, the mean direction of the site is determined by the intersection of the great circles, and an "ellipse of confidence" [32] is calculated. The very poor intersection determination from these great circles for site I (see Table 2) render this site unusable.

(2) Characteristic remanent magnetization (ChRM) directions have been isolated either by thermal demagnetization (site J) or by chemical demagnetization (sites P1 to P6) or by the two techniques (sites C, E, F, G). In the latter case the populations obtained by the two techniques are nearly identical (Fig. 7).

(3) When both ChRM directions and great circles have been determined (sites B and D), we have two estimates of the mean paleomagnetic direction of the site, one by Fisher statistics on the ChRM directions, and the other by great circles intersection. A final mean direction is calculated by averaging these two estimates.

TABLE 2

In-situ (IS) and tilt-corrected (TC) mean directions after chemical and thermal demagnetization

| Sites | $N$ | IS | | TC | | $k$ | $\alpha_{95}$ |
|---|---|---|---|---|---|---|---|
| | | $D$ | $I$ | $D$ | $I$ | | |
| A | 9 | 195 | −4 | 194 | −6 | ($p$ = 15 | $r$ = 9) |
| B | 11 | 213 | −15 | 212 | −12 | 21 | 13.0 |
| C | 14 | 205 | 10 | 205 | −8 | 39 | 6.5 |
| D | 15 | 215 | 10 | 216 | −18 | 22 | 10.0 |
| E | 12 | 193 | −9 | 208 | −29 | 15 | 11.5 |
| F | 16 | 204 | 5 | 206 | −29 | 50 | 5.3 |
| G | 8 | 202 | −7 | 202 | 11 | 4 | 29.5 |
| (I | 7 | 223 | 31 | – | – | $p$ = 35 | $r$ = 9) |
| J | 9 | 211 | −1 | 222 | −29 | 14 | 14.0 |
| P1 | 8 | 210 | 0 | 207 | −40 | 4 | 30.0 |
| P2 | 12 | 207 | −26 | 202 | −39 | 25 | 9.0 |
| P3 | 9 | 210 | −57 | 217 | 13 | 50 | 7.5 |
| P4 | 7 | 242 | −2 | 246 | −39 | 25 | 9.0 |
| P5 | 8 | 235 | 13 | 234 | −5 | 10 | 18.5 |
| P6 | 8 | 232 | 4 | 234 | 26 | 10 | 18.5 |
| Group I (A–J) | 8 $k$ = 45, $\alpha_{95}$ = 8.5 | 205 | −1.5 | 208 $k$ = 24.5, $\alpha_{95}$ = 11.5 | −15 | | |
| Group II (P1–P6) | 6 $k$ = 7.5, $\alpha_{95}$ = 26 | 224 | −11 | 224 $k$ = 6, $\alpha_{95}$ = 29.5 | −15 | | |

$k$, $\alpha_{95}$: parameters of Fisher statistics; $p$, $r$: 1/2 angles of the "ellipse of confidence". The results of the site I are not taken into account in the calculation of the means.

## 5. Results—analysis of the ChRM directions

### 5.1. Group I (gradient sites)

From the results obtained by demagnetization (Table 2), two main conclusions can be drawn. First, the within-site dispersions remain rather high after demagnetization, with always low values of $k$. This contrasts with the results obtained on undeformed beds [8]. Although these directions look like Permian magnetizations, tilt corrections do not produce any improvement in the between site dispersion.

The fact that the regional trend of shortening axis ($Z$) of the strain ellipsoid is subhorizontal in a SSW direction (i.e., more or less parallel to the pretectonic magnetization vector for the undeformed part of the series (Dôme de Barrot)) suggests that the large observed within-site dispersion represents an increase of the initial dispersion, due to a shortening the axis of which is close to the mean of the vectors population.

To check this interpretation, the paleomagnetic data have been compared with a simple numerical model. The modeling conditions are as follows:

(1) The calculations are done on fictitious sets of vectors, each having a Fisherian distribution characterized by the usual statistical parameters ($k$, $\alpha_{95}$) [33].

(2) We apply an axisymmetric strain to the vectors.

(3) The shortening axis is exactly parallel to the initial mean direction of each population.

(4) The vectors behave as material lines under strain (see Appendix 1).

(5) Strain intensity, estimated by the parameter $r = \lambda_1/\lambda_2 + \lambda_2/\lambda_3 - 1$ [41] is incremented by regular coaxial steps until $r = 5$.

(6) At each strain step, the mean of each population is recalculated with its statistical parameters.

These calculations performed on four populations with different initial value of $k$ (14.5, 39, 70, and 125) give the following results:

(1) The mean direction of each population stays parallel to the $Z$ axis.

(2) There is a rapid decay of the Fisher $k$ parameter with increasing $r$ (Fig. 8). In other words, the dispersion of a vectors population under these conditions becomes rapidly large, even for relatively small strain intensities.

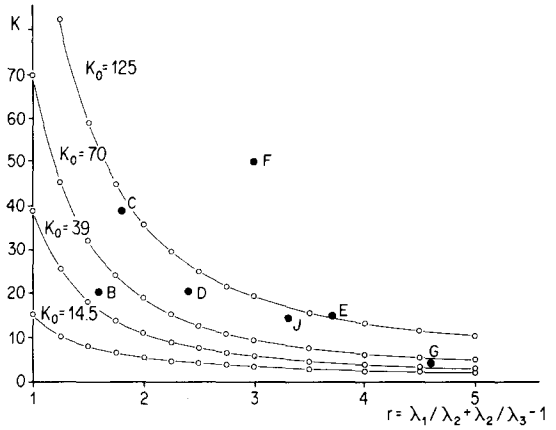



Fig. 8. Curves of modeled behaviour of Fisher's $k$ parameter as a function of strain intensity $r$ [41]. Black dots are real parameters of the group I sites.

With one exception (site F), the real paleomagnetic and structural data for each of the eight sites B to J show a good fit with calculated curves for an initial population of $k$ parameters in the range 50–100, which is the usual value for this kind of rock. In nature, there is probably seldom any true parallelism between pretectonic mean direction of magnetization and shortening axis, and moreover the strain is not uniaxial. Nevertheless, this model may explain the within-site dispersions.

### 5.2. Group II (fold sites)

Although chemical demagnetization allowed us to isolate a stable component for each treated specimen, the dispersions remain very high except for site P3 (Table 2; Fig. 9). In this field, characteristics and intensities of strain are more or less identical from site to site but we have variations in the angles between the strain axes and the bedding plane, in other words, variables angles between the shortening axis and the pretectonic magnetization. It is possible to find, as before, a coherent attitude of paleomagnetic data with respect to strain directions. Comparison of the data shows that in sites P1 and P4 from the southern limb (see Fig. 2)

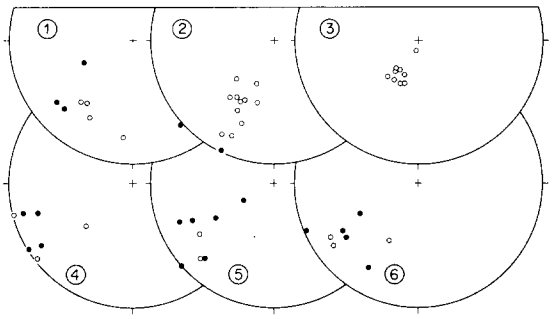



Fig. 9. In-situ ChRM directions obtained by chemical demagnetization in the six group II sites. Same conventions as in Fig. 4.

there exists a large scatter of paleomagnetic data (with $k$ values respectively 4 and 7), as well as in sites P2, P5, P6 from the hinge; this high scatter is consistent with the low angle between the $Z$-axis and the expected mean pretectonic magnetization direction, as observed in the group I sites; in contrast, in site P3 of the northern limb, where this angle is much higher, scatter remains small in the observed state ($k = 50$), but there is a deviation of the whole population away from the shortening axis, towards steep upwards inclinations.

As mentioned above, a simple tilt correction does not bring any improvement in the clustering of the site-mean directions. This is due to the fact that the site-mean directions of the two limbs bypass each other during unfolding (Fig. 10). This type of situation is classically attributed to a syn-

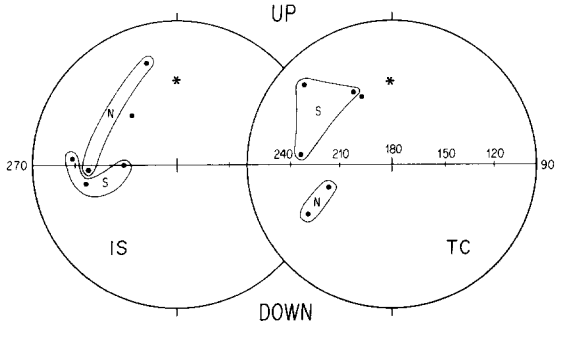



Fig. 10. Site-mean ChRM directions of group II in a vertical stereographic projection. Dots are contoured depending on their southern (S) or northern (N) limb site position. IS: in situ; TC: tilt-corrected; asterisk: recent field axis (reverse polarity). Closed symbols in the southern hemisphere.

tectonic origin of magnetization. However, this cannot be the case here, because the Tertiary age of the deformation would imply a remagnetization field close to the Tertiary field and yet somewhere between the directions of the two limbs, which is not at all the case. Instead the directions are distributed around a Permian field direction. We suggest therefore that strain induces a deviation of the initial magnetization. In a general way, strain has to be considered as an alternative interpretation for magnetic data which show this kind of by-passing upon unfolding.

### *5.3. Conclusions*

(1) When it is possible to isolate a stable ChRM direction in each specimen, these directions are generally scattered within the deformed sites.

(2) This dispersion shows an increase with strain when the shortening direction is close to the paleomagnetic one.

(3) When the angle between the $Z$-axis and the theoretical pretectonic mean paleomagnetic direction is higher, the within-site dispersion seems to remain unaffected, but the whole population is deviated towards the cleavage ($XY$) plane.

Thus the behaviour of the paleomagnetic vectors population studied here is similar to the behaviour of passive vectors under strain. In order to test this hypothesis, and to recover a mean paleomagnetic direction consistent with the Permian direction of Stable Europe, the effects of strain have been removed from each ChRM direction.

## 6. Strain removal

### *6.1. Principle*

The analysis of deformation is a geometrical description of the transformation from an initial state to a deformed state. This description, which is totally independent of the mechanism and the path of deformation, is mathematically decomposed into (a) strain (i.e., changes in relative positions of points, (b) rigid-body rotation, and (c) rigid body translation. The latter is generally not

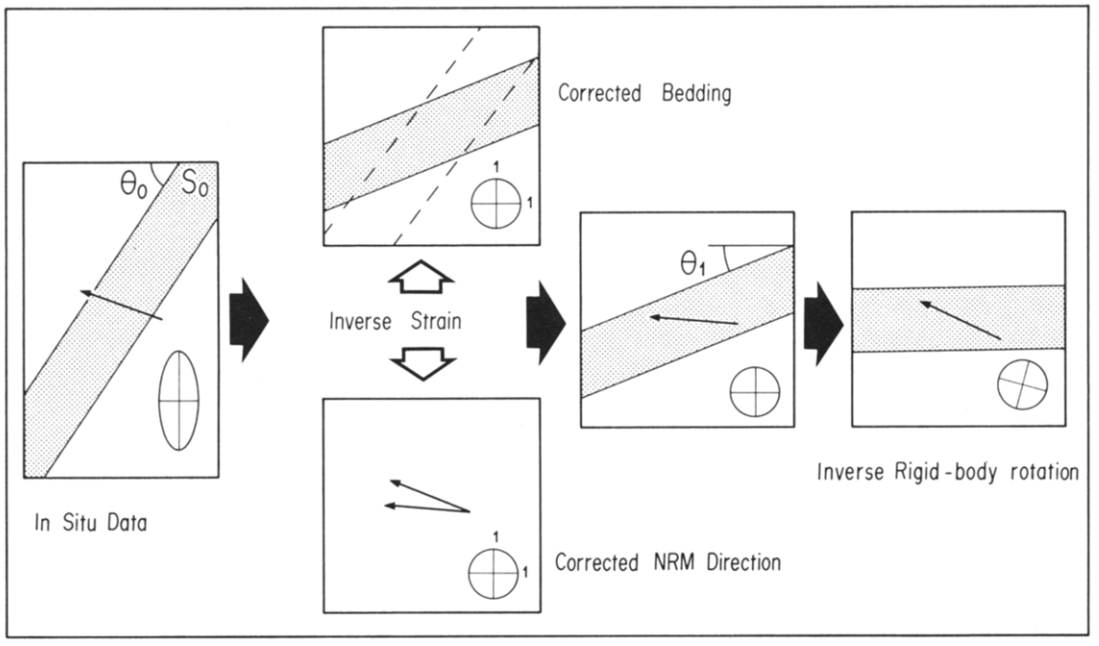



Fig. 11. Schematic sketch of strain removal. From the in-situ paleomagnetic and strain data, both vectors and planes are corrected for strain. Then the inverse rigid-body rotation is done by tilt correction from the unstrained bedding plane.

easy to detect and does not affect paleomagnetic vectors; therefore, it will no longer be considered here. Consequently, total deformation has been removed from the ChRM vectors in two steps (Fig. 11):

(1) First, according to theoretical relationships

TABLE 3

Mean-site and mean-group directions after strain removal and tilt correction from the unstrained bedding plane with the results of the test $k_2 > k_1$

| Sites | $N$ | $D$ | $I$ | $k$ | $\alpha_{95}$ | $k_{SR}/k_{IS}$ | $F$ | Test |
|---|---|---|---|---|---|---|---|---|
| A | 9 | 194 | −11 | ($p=15$ | $r=9$) | | | |
| B | 4 | 213 | −12 | 36 | 13.0 | 1.70 | 4.28 | ? |
| C | 14 | 194 | −8 | 67 | 4.9 | 1.73 | 1.92 | ? |
| D | 7 | 204 | −9 | 82 | 5.0 | 4.45 | 2.69 | + |
| E | 12 | 180 | −11 | 78 | 4.9 | 5.03 | 2.05 | + |
| F | 16 | 196 | −16 | 351 | 2.0 | 7.02 | 1.84 | + |
| G | 8 | 204 | −1 | 44 | 8.5 | 9.78 | 2.48 | + |
| J | 9 | 196 | −16 | 45 | 7.8 | 3.08 | 2.29 | + |
| P1 | 8 | 205 | −7 | 125 | 5.0 | 31.25 | 2.48 | + |
| P2 | 12 | 208 | −11 | 193 | 3.0 | 7.72 | 2.05 | + |
| P3 | 9 | 221 | 6 | 115 | 5.0 | 2.30 | 2.29 | + |
| P4 | 7 | 217 | −9 | 174 | 4.5 | 10.23 | 2.69 | + |
| P5 | 8 | 218 | −0.5 | 96 | 5.7 | 9.60 | 2.48 | + |
| P6 | 8 | 217 | 7 | 107 | 5.5 | 10.70 | 2.48 | + |
| Group I | 8 | 198.0 | −10.5 | 59 | 7.5 | | | |
| Group II | 6 | 214.5 | −2.5 | 65.5 | 8.0 | | | |

$k_{SR}/k_{IS}$: $k$ ratios for strain removed and in-situ directions; $F$: value of the Fisher-Snedecor table at the 5% level for the $2(N-1)$, $2(N-1)$ degrees of freedom.

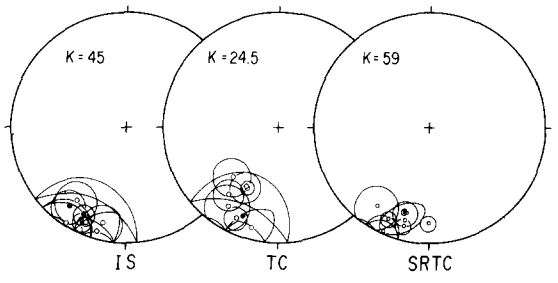



Fig. 12. Group I: site-mean directions with their circles of confidence. IS: in-situ; TC: tilt-corrected; SRTC; after strain removal and tilt correction from the unstrained bedding plane. The Fisher $k$ parameters refer to the between-site dispersions.

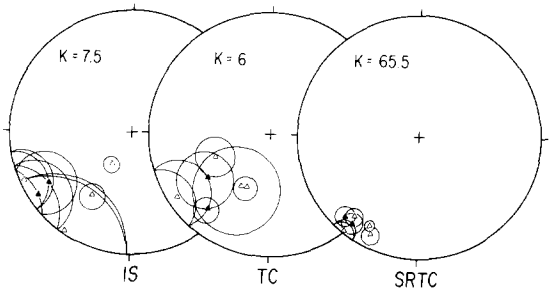



Fig. 13. Group II: site-mean directions with their circles of confidence. Same abbreviations and conventions as in Fig. 12.

between initial and final positions of a passive line under strain (see Appendix 1) the inverse strain tensor $S^{-1}$ has been used to recalculate the initial direction of the ChRM of each specimen, without any rigid-body rotation.

(2) The mean strike and dip of the bedding plane of each site has always been determined by using surfaces with no rheological contrast with the layers where reduction spots were measured. In this case, bedding plane can be considered as a passive plane during deformation and thus, the observed strike and dip of these planes is the result of the superposition of a rotationnal part due to strain and a rigid-body rotation. Consequently, the strain was also removed from the bedding plane (see Appendix 1). The residual strike and dip of the recalculated plane (so-called "unstrained bedding plane") was considered as the part of rigid-body rotation which is associated with the total deformation. Inverse rigid-body rotation of the ChRM vectors was then performed by bringing the recalculated bedding plane of the site to the horizontal with a classical tilt correction (rotation about the strike line, of an angle equal to the dip).

Mean directions of the two groups are given in Table 2 for in-situ and tilt-corrected directions and in Table 3 after strain removal and tilt correction from the unstrained bedding plane. These data for the three stages are illustrated Fig. 12 for group I and in Fig. 13 for group II.

### 6.2. *Effect of strain removal on within-site dispersions*

Strain removal performed on ChRM vectors induces a general displacement of each vector towards the $Z$-axis of the site. Because in most of our sites the $Z$-axis is close to the mean of the paleomagnetic vector population there is a general decrease of the within-site dispersions (Table 3). For the sites distributed along the gradient (group I), the improvement in the clustering, reflected by the $k_2/k_1$ ratios, begins to be significant in a statistical sense after the pencil structure stage (site C), and increases towards the north with the strain intensity. The increase of the $k_2/k_1$ ratios, due to the increase of the inverse strain intensities applied to the directions, is to be related to the general decrease of the $k$ values of observed in-situ ChRM directions along the gradient (group I), reported above. For the six sites of the fold (group II), the improvement in the within-site clustering is always statistically significant but one can notice that the value of the $k_2/k_1$ ratios depends upon the angle between the strain axes and the paleomagnetic direction (site P3: 2.3, site P1: 31.2)

The behaviour of the vector populations under strain removal is illustrated by the reduction of the circles of confidence in Figs. 12 (SRTC) and 13 (SRTC).

Starting from within-site distributions, which were in some cases almost unsuitable for paleomagnetic purposes, strain removal allows us to improve these distributions, and this first step of the treatment may be considered as quite positive.

### 6.3. *Between-site dispersion*

The second step which consists of a tilt correction from the unstrained bedding plane of each

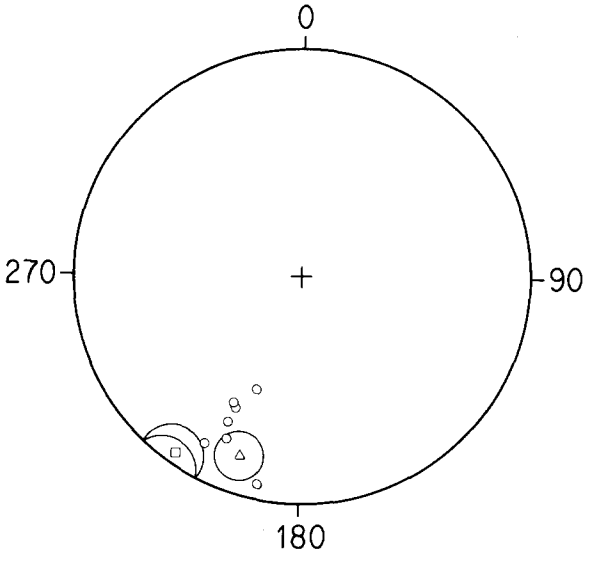



Fig. 14. Group I (triangle) and group II (square) means compared with seven Permian directions (circles) recalculated for the Argentera sites from the poles of Stable Europe [34].

site, gives us further control on the directions recovered by strain removal. Comparison of the site-mean statistics at the three-different stages (Tables 2 and 3) for group I and II shows an improved clustering of the directions after this inverse rigid-body rotation. This improvement is statistically significant with respect to simple tilt correction for group I data ($k$ goes from 24 to 59; $n = 8$) and to both in-situ and tilt-corrected data for group II ($k$ from 7 and 6 to 65; $n = 6$). Furthermore, the mean paleomagnetic directions from the two groups (Fig. 14) show a very good fit with the directions recalculated for the Alpes Maritimes red bed location using the seven most reliable Permian poles of Stable Europe, as reviewed by Van der Voo et al. [34].

This second step of the treatment appears to be positive for the clustering of the between-site distributions as well as for the improvement of the paleomagnetic direction accuracy. It is also the most significant since it is not controlled as strongly as the first step by the shortening axis.

## 7. Summary of the analysis

The observations arising from this study are the following:

(1) The "primary" chemical remanent magnetization acquired by the Permian red beds of the Alpes Maritimes before deformation undergoes an important deviation during the Alpine deformation of these rocks.

(2) This deviation shows a coherence with respect to the strain axes that is reflected by (a) a large scatter of the within-site populations of paleomagnetic vectors, which increases with strain intensity, when the angle between the shortening $Z$-axis and the initial mean direction of the population is low; (b) a deviation of the whole population when this angle is higher.

(3) The deviation which can tentatively be correlated with hematite reorientations during deformation, as shown by texture goniometry [27], shows similarities with macroscopic material lines behaviour. Therefore, an attempt has been made to recalculate pretectonic directions of magnetization by strain removal using the theoretical equations for deformation involving passive lines and planes.

Recalculating each individual ChRM direction by using the inverse strain tensor determined at each sampling site, we obtain positive results at two levels: (a) First, the particular relationships between the $Z$-axis and mean paleomagnetic direction lead to an improvement in the clustering of the population at the site and thus, to an improvement in the site mean direction; (b) after inverse rigid-body rotation by tilt correction from the undeformed bedding plane, the overall mean directions of the two groups of sites show good consistency with known Permian paleomagnetic directions of Stable Europe [34], with good statistical quality parameters.

## 8. Discussion

Strain removal technique applied to remanent magnetization vectors [35,36] has been partly used by Kligfield et al. [20] on their data obtained from the same rocks. These authors observe an important improvement in the within-site clustering of the ChRM directions during strain removal. Based on the fact that under inverse strain, passive markers inevitably move towards the shortening

$Z$-axis, they conclude to a probably large computing effect.

In our opinion, the most important observations are (a) the coherence of the angular deviation of the ChRM vectors with respect to the strain axes directions and intensity, and (b) the agreement with the inferred initial magnetization direction. Therefore, the strain may be considered as responsible for the scatter of the data. Thus, the observed improvement in the within-site distributions during strain removal may no longer be regarded as only a computing effect, even if there is a possible over correction, as far as we cannot be sure of the exact passive behaviour of the vectors. On the other hand, the decomposition of total deformation into strain and rigid-body rotation is a geometrical notion which is not necessary related to the natural processes involved in folding and cleavage development. Therefore, strain removal techniques should include inverse rigid-body rotation. This is only with this second step which has shown positive results in the improvement of between-site distributions, that the treatment may be regarded as fully positive. These results lead us to assume that strain measured in the field gives us a good estimation of the corrections to apply to the ChRM vectors.

In spite of all the points that are to be investigated, these promising results show that it seems possible to use paleomagnetic directions as markers not only of rigid rotations and translations of plates or microplates, but also when plastic deformation occurs.

**Acknowledgements**

We thank Dr. P.R. Cobbold for helpful discussions.

This is a contribution of L.P. C.N.R.S. No. 4661; the work was supported by the French CNRS and Institut National d'Astronomie et de Géophysique through the National Research Project ATP Geodynamique.

## Appendix 1

*Macroscopic material line behaviour under strain [37]*

A material line is defined to join two material points $P_1$ and $P_2$ the vector positions of which are, in a convenient cartesian coordinate system, $r_1$ and $r_2$. Thus the line is defined by the vector difference:

$$l = r_1 - r_2$$

The change in position of points $P_1$ and $P_2$ under a homogeneous strain (neglecting rigid-body rotation and translation) leads to new vector positions $r_{f_1}$ and $r_{f_2}$ related to the initial $r_{i_1}$, $r_{i_2}$ by (f: final, i: initial):

$$r_f = S \cdot r_i,$$

where $S$ is a symmetric (strain) tensor. The material line joining the two points is then defined by the vector difference:

$$l_f = r_{f_1} - r_{f_2} = S \cdot r_{i_1} - S \cdot r_{i_2}$$
$$= S \cdot (r_{i_1} - r_{i_2}),$$
$$l_f = S \cdot l_i$$

So the change in length and coordinates of a material line is directly related to its initial length and coordinates and to the strain tensor $S$.

With respect to the chosen coordinate system, the cartesian components of $S$ are (using the summation convention):

$$S_{ij} = a_{ik} \cdot a_{jm} \cdot S'_{km} \qquad (i, j, k, m = 1, 2, 3)$$

where $S'$ is the matrix of the three eigenvalues of $S$ and is of the form:

$$S' = \begin{bmatrix} 1 + e_1 & 0 & 0 \\ 0 & 1 + e_2 & 0 \\ 0 & 0 & 1 + e_3 \end{bmatrix}$$

$e_i$: elongation along each axis and $a_{ij}$ are the direction cosines of the three eigenvectors in the chosen reference frame.

Knowing the coordinates of the vector defining a material line in the deformed state, the initial coordinates can be calculated in the same way by using the inverse strain tensor:

$$l_i = S^{-1} \cdot l_f$$

where

$$S_{ij}^{-1} = a_{ik} \cdot a_{jm} \cdot S'^{-1}_{km} \qquad (i, j, k, m = 1, 2, 3)$$

*Material plane behaviour [38,39]*

We define $n_i$ ($n_f$) as the normal to the initial (final) plane and $l_i$ ($l_f$) as a vector within the initial (final) plane.

$l$ and $n$ are at right angle, thus their scalar product is:

$$n_i \cdot l_i = n_f \cdot l_f = 0$$

from equations above, we have $l_f = S \cdot l_i$ so:

$$n_i \cdot l_i = n_f \cdot S \cdot l_i$$

$$n_i = S^T \cdot n_f$$

where $S^T$ is the transpose of $S$; $S$ is a symmetrical tensor, therefore:

$$n_i = S \cdot n_f$$

So initial coordinates of the normal to a passive plane are recovered by multiplying the strain tensor to the final (observed) normal coordinates.

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